The Cognitive Resolution Protocol:

An Application-Layer Standard for Multi-Threaded Cognitive Perception in AI Systems

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Abstract

Current AI systems treat observation as classification: a single model processes an input and produces a single interpretation. This paper argues that what AI lacks is not better execution but richer perception—the kind of multi-threaded, context-dependent, emotionally informed interpretation that characterizes human cognition. We propose the Cognitive Resolution Protocol (CRP), an application-layer standard for multi-threaded cognitive perception in AI systems.

CRP is inspired by Global Workspace Theory (GWT) from neuroscience but extends it in five critical ways. First, we separate perception into a two-tier architecture: sensory systems that produce rich, general-purpose contextual output, and cognitive modules that interpret those outputs through specialized lenses. Second, we elevate memory from a queryable module to infrastructure—a substrate that primes every cognitive module with contextually relevant history before interpretation begins, analogous to how human memory is distributed through the same neural tissue that performs perception. Third, we introduce affective pre-processing: a fast, sub-cognitive layer that generates emotional valence before conscious classification, biasing cognitive module priorities in real-time. Fourth, we propose twin-specific priority learning: module priority weights learned from observing individual human behavior. Fifth, we formalize an instinct mechanism: modules can signal partial relevance detection without producing a full interpretation, creating persistent signals that resolve across observation cycles.

CRP is published as an open standard. The protocol defines a universal packet format, five response types (assertion, query, null, override, resolve), a ten-step workspace cycle, and a volatile RAM working memory model with channel-dependent decay. A reference implementation with nine modules is operational in a production digital twin system.

Keywords: cognitive architecture, perception protocol, Global Workspace Theory, affective computing, working memory, digital twins, open standard, application-layer protocol, multi-agent cognition

1. Introduction

The automation of knowledge work has followed a consistent trajectory: observe what the human does, record the steps, replay them. From early robotic process automation (RPA) through modern AI-powered browser agents, the fundamental paradigm remains observation-as-classification. An AI watches a user interact with software, labels the activity, compiles a sequence of steps, and executes them mechanically.

This paradigm has produced useful tools but fundamentally limited ones. When a senior behavioral health clinic administrator processes payroll, she is not executing a script. She carries working knowledge: this staff member was on FMLA last month, that insurance reimbursement typically lags by three weeks, this particular payer denies first-pass claims on certain CPT codes. She notices anomalies that would be invisible to a step-replay system. She adjusts her approach based on context that exists nowhere in the recorded workflow.

The gap between human task performance and automated task replay is not a gap in execution capability—modern AI can click buttons and fill forms with high accuracy. The gap is in perception. The human perceives the task environment through multiple simultaneous interpretive lenses. The AI system classifies it through one.

This paper proposes an architecture and open protocol that bridges this gap. The Cognitive Resolution Protocol (CRP) is an application-layer standard (OSI Layer 7) that defines how an AI system's internal cognitive modules collectively interpret a single observation to produce meaning. CRP sits in the same architectural layer as HTTP and DNS, but where those protocols define how systems communicate with each other, CRP defines how a system communicates within itself—how the components of an AI mind coordinate to achieve perception.

1.1 The Flattening Problem

Consider a single screenshot of an EHR billing screen. A classification-based system produces one interpretation: "user is on the billing module, viewing patient #4412's claims." This is accurate but impoverished. A human looking at the same screen simultaneously processes:

Task context: "This is step 4 of my payroll reconciliation workflow."

Anomaly detection: "These hours look unusual for this staff member."

Compliance awareness: "This payer usually requires modifier 25 on this code."

Privacy assessment: "Patient name is visible; I should be careful about screen sharing."

Affective response: "Red indicators on three claims—something feels wrong."

Temporal awareness: "It's 4:55 PM and billing closes at 5. This is urgent."

Each of these is a valid interpretation of the same visual input, processed in parallel, with different priorities, different persistence requirements, and different downstream implications. Current AI systems collapse this multi-dimensional perception into a single label. We call this the flattening problem.

1.2 CRP in the Protocol Landscape

The agentic AI ecosystem has developed two major communication protocols: Anthropic's Model Context Protocol (MCP), which standardizes how agents access external tools and data sources, and Google's Agent-to-Agent Protocol (A2A), which enables peer-to-peer agent communication and task delegation. Both address inter-agent communication: how AI systems talk to each other and to the external world.

CRP addresses a different and complementary problem: intra-agent cognition. How does a single AI system's internal architecture process an observation to produce meaning? MCP defines how an agent accesses tools. A2A defines how agents talk to each other. CRP defines how an agent thinks about what it perceives. It is the perception layer of the agentic AI stack.

1.3 Contributions

This paper makes the following contributions:

1. A two-tier perception architecture separating sensory systems (which produce rich, general-purpose output) from cognitive modules (which interpret through specialized lenses), mirroring the biological separation between sensory transduction and cortical interpretation.
2. Memory as infrastructure, not module: memory is elevated from a queryable service to a substrate that primes all cognitive modules with contextually relevant history before each broadcast cycle, with on-demand retrieval for targeted queries and post-cycle formation for new memories.
3. Affective pre-processing: a fast, sub-cognitive layer that generates emotional valence from sensory input before cognitive classification, biasing module priorities in a manner analogous to the amygdala's role in human threat detection.
4. A volatile RAM working memory model with channel-dependent decay rates, where memories generated by different cognitive processes persist for different durations based on their function, emotional weight, and outcome significance.

5. An instinct mechanism: cognitive modules can signal partial relevance detection ("something about this feels important") without producing a full interpretation, creating persistent signals that resolve across observation cycles when new data arrives.
6. A formal protocol specification published as an open standard: universal packet format, five response types (assertion, query, null, override, resolve), ten-step workspace cycle, and transport-agnostic design.
7. Twin-specific priority learning: module priority weights that are not hardcoded but learned from individual user behavior, capturing how different humans allocate cognitive resources under different conditions.

2. Related Work

2.1 Global Workspace Theory

Global Workspace Theory, proposed by Baars (1988) and extended by Dehaene and colleagues as Global Neuronal Workspace Theory (GNWT), provides a framework for understanding conscious access. In GWT, specialized processing modules operate largely in parallel and unconsciously. When a module's output is selected for "broadcast" to the global workspace, it becomes available to all other modules simultaneously. This broadcast-and-response cycle is the mechanism by which coherent interpretation emerges from distributed processing.

Critically, GWT does not propose that the broadcast produces a single interpretation. The broadcast is a query—raw sensory data or a candidate interpretation sent to all modules simultaneously. Each module responds independently: threat assessment, memory retrieval, motor planning, emotional evaluation. These responses may conflict, may be partial, or may generate further queries. The global workspace aggregates these responses and selects the next broadcast based on salience and priority.

Recent work has applied GWT principles to AI architecture. Nakanishi et al. (2025), publishing in Frontiers in Robotics and AI, identified three functional advantages of GWT's selection-broadcast cycle for real-time AI systems: dynamic thinking adaptation, experience-based adaptation, and immediate real-time adaptation. Chateau-Laurent et al. (2025) demonstrated that GNWT-based architectures outperform LSTM and Transformer baselines in causal and sequential reasoning tasks. VanRullen and Kanai proposed explicit workspace architectures with specialized generative modules and individual latent spaces. However, existing implementations typically produce a single output per cycle, discarding the parallel multi-modal responses that give human cognition its richness.

2.2 Cognitive Memory Architectures

Classical cognitive architectures such as ACT-R (Anderson, 1983) and Soar (Laird, 2012) formalized the separation between declarative and procedural knowledge, with mechanisms for retrieval based on activation levels that decay with time and strengthen with use. Modern AI memory systems such as Mem0, MemoryBank, and the Memoria framework implement similar principles, using exponential decay functions to manage memory relevance over time.

The FiFA framework (2025) introduced privacy-aware memory with selective retention, combining temporal hygiene with importance estimation and reflection-driven consolidation. However, these systems treat all memories as uniform objects with a single decay function. They do not distinguish between memories generated by different

cognitive processes—a task-relevant observation and a peripheral aesthetic impression decay at the same rate. Furthermore, they treat memory as a module to be queried rather than as infrastructure that shapes all processing.

2.3 Affective Computing and Digital Interface Perception

Research in affective computing has established that emotional responses to stimuli are processed through fast subcortical pathways (notably the amygdala) before conscious classification occurs. Damasio's somatic marker hypothesis demonstrated that patients with damage to emotional circuits struggle with rational decision-making, establishing that affect is not separate from cognition but integral to it.

In human-computer interaction, color psychology research has shown that interface colors produce measurable affective responses: red activates alertness and urgency, blue produces trust and calm, green signals safety and completion. These responses occur pre-consciously and influence subsequent cognitive processing. The application of affective pre-processing to AI perception of digital interfaces (rather than human faces or voices) is, to our knowledge, novel. No current AI observation system generates emotional valence from screen content as a perception input.

2.4 Agent Communication Protocols

The agent protocol landscape has consolidated around two primary standards. Anthropic's Model Context Protocol (MCP), governed by the Linux Foundation since December 2025, standardizes how agents connect to external tools, databases, and APIs. Google's Agent-to-Agent Protocol (A2A), also under Linux Foundation governance since June 2025, enables horizontal communication between autonomous agents. Both address inter-agent communication. Neither addresses intra-agent perception: the question of how a single agent's internal cognitive architecture processes an observation.

2.5 Gap Analysis

The literature reveals a clear gap at the intersection of these fields. GWT-inspired AI architectures exist but flatten the multi-modal response cycle into single-output classification. Cognitive memory architectures implement decay but treat all memories uniformly and position memory as a queryable module rather than perceptual infrastructure. Affective computing detects human emotions but does not apply affect to AI perception of digital interfaces. Agent protocols define inter-agent communication but not intra-agent cognition. No existing system combines a two-tier perception hierarchy, memory as infrastructure, affective pre-processing, channel-dependent memory decay, instinct mechanisms, and a formal open protocol specification.

3. Proposed Architecture

We propose a layered cognitive perception architecture organized into five processing stages. From fastest to slowest: (0) sensory transduction, (0.5) memory context priming, (1) affective pre-processing, (2) cognitive broadcast and response, and (3) memory formation. These stages operate in sequence within a single observation cycle but the architecture supports concurrent cycles across observations.

3.0 Layer 0: Sensory Systems

The first major architectural contribution of CRP is the separation of perception into sensory systems and cognitive modules. This mirrors the biological separation between sensory organs and cortical interpretation. The human ear does not interpret—it transduces sound waves into electrical signals, decomposes them into frequency components, and passes them to higher-order systems. The auditory cortex interprets. The vestibular system in the same organ provides spatial orientation. One organ, multiple outputs, consumed by different cognitive consumers.

CRP sensory systems follow this principle: they receive broadly, process at multiple levels of abstraction, and produce rich contextual output that cognitive modules consume in ways the sensory system did not anticipate. A sensory system is general-purpose. A cognitive module is specialized.

Sensory System	Input	Output	Sub-Components
Visual Sense	Screenshot / frame	Color distribution, spatial layout, element density, icon/badge detection, modal presence, change from prior frame	Color analyzer, layout classifier, element detector, change detector
Textual Sense	Extracted text, OCR output	Entity extraction, pattern recognition, information density, language register, structural analysis (form vs. table vs. free text)	Entity recognizer, structure classifier, density estimator, register analyzer
Temporal Sense	Timestamps, interaction events	Time-on-page, click velocity, navigation rhythm, session duration, time-of-day context	Pace tracker, rhythm analyzer, session profiler
Contextual Sense	Window metadata, URL, process info	App identity, workspace/project context, navigation state, tab topology	App classifier, workspace detector, state tracker

Auditory Sense	Microphone input (when available)	Speech transcription, speaker identification, ambient noise level, conversation vs. monologue, emotional prosody	Transcriber, speaker diarizer, prosody analyzer

Critically, sensory systems are designed to be general and rich, not narrow and opinionated. The Textual Sense does not decide whether text contains PHI—it extracts entities, detects patterns, estimates sensitivity, and reports the language register. The Safety cognitive module consumes that output and makes the PHI determination. The Task cognitive module consumes the same output and matches workflow patterns. The Anomaly module consumes it and checks for deviations from expectations. One sensory output, multiple cognitive interpretations.

This separation provides two engineering advantages. First, adding new cognitive modules does not require new sensory infrastructure—a new Billing module for a finance vertical simply consumes the existing Textual and Contextual senses. Second, sensory systems run locally on the user's device (they are fast heuristics, no LLM required), while cognitive modules can run in the cloud, with the compact sensory context transmitted rather than raw screenshots.

3.0.5 Layer 0.5: Memory Context Priming

The second major architectural contribution is elevating memory from a queryable module to infrastructure. In human cognition, memory is not a separate organ that other brain regions query—it is distributed through the same neural tissue that performs perception and reasoning. The hippocampus does not wait for a request; it automatically primes relevant episodic memories when contextual cues arrive. Walking into a familiar room triggers recall before any conscious effort to remember.

In CRP, after sensory systems produce their outputs but before cognitive modules process them, the memory substrate assembles a context frame: a compact packet of relevant memories selected based on the current sensory picture. The Contextual Sense reports "we are in CheckpointEHR, CP Central dashboard." Memory immediately retrieves: the last several observations from this application, known workflow patterns, unresolved instinct signals associated with this context, and recent anomaly flags. This context frame is attached to the broadcast packet—every cognitive module receives it automatically.

Memory serves three distinct roles within each CRP cycle:

Pre-load (context priming): Before the broadcast, memory curates a contextually relevant subset based on sensory outputs. This is automatic, not requested. Cognitive modules receive pre-loaded context without asking for it.

On-demand retrieval: During the broadcast cycle, a cognitive module may need something specific that was not in the pre-loaded context. The Anomaly module detects an unusual pattern and queries: "Have I seen this specific billing code reject before?" Memory performs a targeted semantic search, scoped by the full sensory context, and returns a relevance-scored subset—not the full memory.

Post-cycle formation: After the cycle completes, memory decides what to store. The salience score determines encoding strength. The affect context determines the memory channel and decay rate. High-salience observations with strong emotional valence are stored strongly in slow-decay channels. Routine observations go into fast-decay peripheral channels. Overrides are stored permanently for audit.

The retrieval mechanisms support three search types: episodic search (contextual matching—same app, similar screen, similar time of day), semantic search (embedding similarity—conceptually related memories regardless of surface features), and pattern search (temporal sequence matching—recurring workflows and behavioral patterns).

3.1 Layer 1: Affective Pre-Processing

Before any cognitive classification occurs, each observation passes through an affective pre-processor that generates four signals from the sensory system outputs:

Signal	Description	Latency Analogue	Effect on Cognitive Module Priorities
Valence	Positive / negative / neutral emotional tone	Amygdala fast path (~50ms)	Negative valence boosts safety, compliance, anomaly modules
Arousal	High / low cognitive activation energy	Sympathetic nervous system	High arousal boosts task focus; suppresses background learning
Familiarity	Known pattern / novel stimulus	Hippocampal matching	Novel stimuli boost anomaly and learning modules
Urgency	Immediate action needed / can wait	Prefrontal priority assessment	High urgency boosts task module; suppresses peripheral processing

In v0.1.0 of CRP, the affective pre-processor used heuristics on raw screenshots. In this revised architecture, affect draws from all five sensory system outputs: Visual Sense's color distribution provides valence (red dominance → negative, green → positive). Textual Sense's entity density provides arousal (dense data screens → high cognitive load). Temporal Sense's pace data provides urgency (long time-on-page → either deep work or being stuck). Contextual Sense's app familiarity provides the familiarity dimension. This produces richer affect signals because the inputs themselves are richer.

The affect computation must complete in under 100ms to precede the cognitive broadcast. Because it draws from pre-computed sensory outputs rather than raw pixels, this constraint is achievable with pure heuristics—no LLM required.

3.2 Layer 2: Cognitive Broadcast and Response

Following sensory transduction, memory priming, and affective pre-processing, the observation is broadcast to all cognitive modules simultaneously. The broadcast packet contains: the raw observation identifier, sensory system outputs, the pre-loaded memory context frame, and the computed affect state. Each cognitive module processes the broadcast independently and returns one of five response types:

Assertion: A confident interpretation. "This is step 4 of the payroll workflow. Confidence: 0.85." Immediately useful, contributes to salience computation.

Query: A partial relevance detection—the instinct mechanism. "Something about these hours looks unusual; I need historical comparison data." The module senses relevance but cannot yet articulate a full interpretation. Query responses persist in volatile RAM across cycles until resolved.

Null: The observation is not relevant to this module. No response generated, minimal resources consumed. Most modules return null on most observations.

Override: A high-priority signal that preempts all other processing. "PHI detected—suppress cloud upload immediately." Safety and privacy modules can issue overrides that take effect before other modules complete.

Resolve: The answer to a previously issued query. "The hours anomaly from three cycles ago is explained—this employee switched to a different shift schedule last week." Resolves an instinct signal and optionally promotes the resolution to an assertion.

3.2.1 Cognitive Modules Are Consumers, Not Sensors

A critical design principle: cognitive modules do not perform raw perception. They interpret the outputs of sensory systems. This is the distinction between the cochlea (sensory transduction) and the auditory cortex (interpretation). The human auditory system does not have one component—it is a hierarchy of sub-components: the cochlea converts sound waves to electrical signals, the auditory nerve decomposes frequencies, brainstem nuclei perform sound localization, and the auditory cortex performs pattern recognition. The vestibular system shares the same organ but serves a completely different cognitive function (spatial orientation).

Similarly, CRP sensory systems are hierarchies of sub-components that produce rich, general-purpose output. Cognitive modules consume these outputs for specialized purposes. Adding a new cognitive module requires no new sensory infrastructure. A Billing module for a finance vertical consumes the same Textual Sense and Contextual Sense outputs that a Clinical module uses in healthcare.

3.2.2 Module Taxonomy

Cognitive modules are organized by function. The reference implementation includes:

Module	Consumes	Function	Can Override
Safety	Textual Sense (entities, sensitivity), Visual Sense (credential fields), Contextual Sense	PHI/PII detection, sensitive data protection, credential exposure	Yes
Task	Contextual Sense (app, URL), Textual Sense (patterns), Temporal Sense (session phase), Memory	Skill/workflow identification, step tracking, task switch detection	No
Anomaly	All senses + Memory + Affect	Deviation detection, instinct generation, expectation matching	No
Learning	Temporal Sense (patterns), Task (sequences), Memory (frequencies)	Recurring behavior identification, pattern promotion to knowledge	No
Intent	Task (workflow), Temporal Sense (dwell time), Textual Sense (reading/typing)	Goal inference distinct from workflow identification	No
Compliance	Textual Sense (codes, register), Contextual Sense (app domain)	Regulatory rule checking, standard application	No
Social	Contextual Sense (app classification), Auditory Sense (conversation detection)	Team/collaboration context detection	No

This taxonomy is extensible. Vertical-specific deployments add domain modules: Clinical for healthcare (treatment modality detection, diagnostic code recognition), Billing for finance (transaction pattern recognition, risk assessment), Legal for law (clause detection, precedent matching). The sensory systems remain unchanged; only the cognitive interpretation layer specializes.

3.2.3 Direct Module-to-Module Communication

A departure from simplified GWT implementations is that cognitive modules communicate directly with each other, not only through the central workspace. This

mirrors neuroscience findings that cortical modules have direct anatomical connections and coordinate without prefrontal mediation.

In CRP, the Task module directly primes the Anomaly module with expectations ("we are in step 3 of payroll; expect the date range selector next"). The Anomaly module directly queries Memory for historical comparison. The Safety module directly commands redaction of sensitive regions. These inter-module communications happen in parallel with the broadcast-response cycle and may span multiple cycles.

3.2.4 Twin-Specific Priority Learning

Module priority weights are not hardcoded. They are learned from observing how the individual human allocates attention across tasks. Some humans under time pressure become hyper-focused (task module dominates, anomaly detection suppressed). Others become hyper-vigilant (anomaly module amplifies). Some are naturally detail-oriented; others are pattern-focused.

The twin learns these priority weights through observation. By correlating context (time pressure, task type, error frequency) with user behavior (what they attend to, what they skip, what they double-check), the system builds a personalized priority map that shifts based on detected cognitive state.

3.3 Volatile RAM (Working Memory)

Between permanent long-term knowledge (the twin's identity, skills, and learned patterns) and transient sensory input (the current observation), volatile RAM provides a working memory layer distinguished by channel-dependent decay rates.

In human cognition, a street-crossing scenario illustrates the principle. You observe a blue Honda Civic approaching from the left. Your brain captures make, model, color, speed, and distance. But the working memory representation that guides behavior is compressed: "threat from left, do not cross." After successfully crossing, the abstract lesson is reinforced in long-term memory. The specific detail ("blue Honda Civic") persists for minutes, then fades. But if you had been car shopping, that detail would persist longer—it was tagged by the background goal system as relevant.

Volatile RAM implements this through channel-tagged memories with independent decay functions:

Memory Channel	Decay Rate	Example	Persistence
Task (immediate)	Rapid: ~minutes	"Currently on payroll step 4"	Until task completes or context switches
Anomaly (flagged)	Medium: ~hours	"Sarah's hours looked unusual"	Until investigated or explicitly dismissed
Instinct (unresolved)	Slow: ~session	"Something about that screen felt off"	Until resolved by new data or naturally decayed
Learning (pattern)	Consolidation cycle	"This payer always denies first-pass"	Promoted to long-term if reinforced; else dropped
Peripheral (incidental)	Fastest: ~seconds	"The login screen had a new banner"	Decays immediately unless another module queries it
Override (audit)	Very slow: ~24 hours	"PHI was detected and upload was blocked"	Retained for audit trail

Three criteria govern promotion from volatile RAM to long-term memory: (1) frequency of recall—if a volatile memory is accessed multiple times across cycles, it strengthens; (2) emotional weight—memories tagged with high affective arousal persist longer and promote more easily; and (3) outcome significance—if a volatile memory was associated with an error, correction, or behavioral change, it promotes. This mirrors the neuroscience finding that emotionally charged and consequential experiences consolidate more readily into long-term memory.

3.4 Concurrent Twin Reasoning During Execution

Existing automation systems treat observation and execution as sequential phases. CRP proposes that the perception pipeline continues operating during execution, providing real-time cognitive partnership.

Pre-step reasoning: Before each execution step, the twin consults volatile RAM and the current sensory context. "Step 3 says select the standard date range, but the Temporal Sense indicates this is month-end, so I should use the full-month view."

In-step monitoring: During execution, sensory systems continue capturing. If the result of an action does not match expectations, the Anomaly module fires.

Post-step reflection: After each step, the twin compares actual to expected. Discrepancies update volatile RAM and may generate instinct signals.

Adaptive override: If accumulated anomaly signals cross a threshold, the twin pauses execution. "I was about to submit this claim, but the modifier doesn't match what this payer typically requires. Should I proceed?"

4. The Instinct Mechanism

Perhaps the most novel aspect of CRP is the instinct mechanism—the ability for a cognitive module to signal "something about this observation feels relevant, but I cannot yet articulate what." This corresponds to the human experience of intuition: a vague sense that something is important, prior to the ability to explain why.

Technically, an instinct signal is a Query response from a module that detected a partial pattern match but lacked sufficient context to produce a full Assertion. In networking terms, it is an ARP request that has been sent but not yet answered—the system knows something relevant exists but has not resolved it.

Instinct signals are stored in volatile RAM's instinct channel with a medium decay rate. They persist across observation cycles, waiting for new data that might resolve them. When a subsequent observation provides the missing context, a Resolve response is generated, the instinct is cleared, and the resolution may be promoted to an Assertion. If no resolution arrives, the instinct fades naturally.

This mechanism is critical for the kind of "background awareness" that characterizes experienced workers. A senior administrator might notice something subtly off about a billing screen and only realize hours later—when reviewing a different patient's file—what the discrepancy was. The instinct persisted subconsciously until the resolving context arrived.

5. Protocol Specification Summary

CRP is specified as an application-layer protocol, transport-agnostic by design. The reference implementation runs over in-process function calls (desktop app, sub-50ms per cycle) but the same protocol operates over HTTP, gRPC, or WebSocket for cloud-hosted deployments.

5.1 Packet Format

The CRP packet is the universal message format for all inter-module communication. Every broadcast, response, query, override, instinct, and resolution uses this structure. Key fields: packet_id (unique identifier), cycle_id (groups all packets from one observation), type (broadcast/direct/response/override/resolve), source_module, target_module ('*' for broadcast), observation_id, twin_id, payload (module-specific data), affect (the four-signal affect context), priority (0.0-1.0), ttl (time-to-live in ms; -1 = persist until resolved), response_type (assertion/query/null/override/resolve), memory_channel, and decay_hint (initial strength, decay function, half-life, promote threshold, emotional weight).

5.2 The Workspace Cycle

Each observation is processed through a ten-step cycle:

8. Sensory transduction: all sensory systems process the raw observation in parallel, producing structured outputs.
9. Memory context priming: the memory substrate assembles a context frame from sensory outputs—recent history, known patterns, unresolved instincts.
10. Affective pre-processing: affect module generates valence, arousal, familiarity, urgency from sensory outputs and memory context.
11. Priority adjustment: module priorities are adjusted based on affect state + twin-learned weights + active priority shifts.
12. Cognitive broadcast: the observation (with sensory context, memory context, and affect) is broadcast to all cognitive modules simultaneously via Promise.allSettled with per-module timeouts.
13. Override processing: any Override responses take immediate effect (e.g., blocking cloud upload). If overrides exist, the cycle short-circuits.
14. Salience computation: weighted sum of all non-null responses, incorporating priority, confidence, and twin-specific module weights.
15. Consciousness gate: if salience exceeds the twin's consciousness threshold, the observation is flagged for active attention. Most observations stay below threshold—the twin perceives them but does not consciously attend.
16. Memory formation: responses are stored in volatile RAM according to their memory channel and decay hints. High-salience, high-affect memories are stored strongly; peripheral observations fade quickly.
17. Instinct resolution: pending Query responses from previous cycles are checked against new data. Resolved instincts generate Resolve responses.

6. Implementation Context: ERA AI

CRP is implemented within ERA AI, a digital twin platform for behavioral health operations. The reference implementation comprises a Chrome extension and Electron desktop application with nine cognitive modules (including the affect pre-processor), a volatile RAM implementation with six memory channels, a reflex gate for pre-workspace filtering, and a priority learner that adapts module weights from user behavior. The system processes observations in under 50ms per full CRP cycle when running locally.

The current deployment supports three execution modes: fully local (all modules in the Electron process, maximum privacy, no network dependency), hybrid (sensory systems and safety modules local, cognitive modules in the cloud via HTTP), and fully cloud (for web-only users without the desktop app, where the Chrome extension sends observations to a cloud CRP endpoint). This flexibility validates the transport-agnostic design of the protocol.

6.1 Reflex Gate

Below the CRP workspace, a reflex gate provides sub-millisecond filtering. Reflexes bypass the workspace entirely: idle detection (no screen changes), duplicate detection (same window title and content), blocklist enforcement (password managers, sensitive apps), and throttling (static pages don't need frequent processing). Reflexes are analogous to spinal reflexes in the human nervous system—fast, automatic, and not requiring conscious processing.

6.2 Optimal Module Count

In human cognition, GWT research suggests the global workspace integrates input from dozens of specialized modules, but only 4-7 competing signals typically reach conscious access per cycle. The rest respond with null or sub-threshold contributions. CRP operates similarly: adding modules does not linearly increase cycle time because null responses are essentially free, all modules run in parallel (cycle time equals the slowest active module, not the sum), and most frames never reach the workspace due to reflex gate filtering.

Empirically, 9-15 cognitive modules is the sweet spot for production deployments: enough diversity for rich perception, few enough for meaningful salience computation and debuggability. Fewer than 5 modules produces a pipeline, not perception—the cross-module interactions that generate emergent understanding require sufficient cognitive diversity.

7. Discussion

7.1 Senses as Infrastructure, Cognition as Application

The two-tier separation mirrors a principle from systems architecture: infrastructure should be general and stable; applications should be specialized and extensible. Sensory systems are the infrastructure layer—they change rarely and serve all cognitive consumers. Cognitive modules are the application layer—they specialize by domain, multiply per vertical, and can be deployed independently.

This separation also enables a natural deployment boundary. Sensory systems run locally (they are fast heuristics on the raw observation). Cognitive modules can run anywhere—locally for maximum privacy, in the cloud for resource-constrained devices, or in a hybrid configuration. The compact sensory context (structured data, not raw pixels) is the interface between local and cloud processing.

7.2 Memory as Medium, Not Module

The elevation of memory from module to infrastructure is perhaps the most architecturally significant revision in CRP v0.2.0. In v0.1.0, memory was listed alongside task, anomaly, and privacy as a peer module that received broadcasts and responded to queries. This created a chicken-and-egg problem: cognitive modules needed memory context to interpret observations, but memory needed to receive the broadcast to know what context to provide.

The revised architecture resolves this by giving memory three distinct temporal roles: pre-load (before broadcast), on-demand retrieval (during broadcast), and post-cycle formation (after broadcast). Memory is not a participant in the conversation—it is the medium in which the conversation happens. Every cognitive module is partially a memory system: it reads from memory-primed context and writes to memory through its response packets. The dedicated memory substrate is the custodian—managing storage, decay, promotion, retrieval indexing, and garbage collection.

7.3 Why Priority Cannot Be System-Defined

A hardcoded priority system ("safety always wins, task is second") would fail to capture the diversity of human cognitive styles. Some workers are methodical checkers; others are pattern-matching skimmers. Some become more careful under stress; others become more impulsive. The twin's value as a cognitive partner depends on mirroring the specific human it serves—not an idealized average. Twin-specific priority learning is therefore not an optimization but a requirement.

7.4 CRP as Open Standard

CRP is published as an open standard because perception architecture should be shared infrastructure. HTTP did not become valuable because one company owned it—it became valuable because everyone could build on it. We believe the same is true for cognitive perception protocols.

Any team building AI agents, digital twins, autonomous systems, or human-AI collaboration tools can implement CRP. The protocol is transport-agnostic: it works over HTTP, gRPC, WebSocket, or in-process function calls. Sensory systems and cognitive modules implement a standard interface (register, initialize, process, shutdown). The workspace cycle is language-agnostic. Our reference implementation is TypeScript; a Python implementation would be straightforward.

7.5 Limitations and Honest Assessment

The reference implementation has been tested with a small number of users at a single behavioral health organization. No formal benchmarks have been completed on the sensory-cognitive separation or memory priming mechanisms. The two-tier architecture described in this paper represents an evolution from v0.1.0 (where all modules were peers) to v0.2.0 (with sensory/cognitive separation and memory as infrastructure). The v0.2.0 architecture is specified but not yet fully deployed.

Specific open questions include: (1) whether the sensory context frame is compact enough for efficient cloud transmission while retaining sufficient richness for cognitive interpretation; (2) whether memory pre-loading introduces bias that narrows cognitive interpretation rather than enriching it; (3) whether twin-specific priority learning converges to useful weights with the limited training signal available from observation alone; (4) whether the instinct mechanism produces genuine value or primarily generates noise; and (5) what the optimal number of sensory sub-components is per sense, and whether sub-component decomposition should itself be configurable per deployment.

8. Conclusion

The AI automation industry has optimized for execution fidelity—how accurately can a system replay recorded actions? This paper argues that the frontier is not better execution but richer perception. When a digital twin can perceive its environment through multiple simultaneous interpretive lenses, remember what matters while releasing what doesn't, feel the emotional tone of a situation before classifying it, and reason concurrently during execution, it transitions from an automation tool to a cognitive partner.

The Cognitive Resolution Protocol provides a formal, open, implementable standard for building such systems. By separating sensory transduction from cognitive interpretation, elevating memory from module to infrastructure, and formalizing affect, instinct, and priority learning as first-class protocol features, CRP offers a foundation that is general enough for any AI perception system yet specific enough to implement and test.

The goal is not to replicate human consciousness. It is to replicate the cognitive structures that make experienced human workers effective: the ability to perceive multiple things simultaneously, to feel before thinking, to hold context that matters while releasing context that doesn't, and to reason while acting. CRP is the protocol for AI that doesn't just process—it perceives.

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